

Sensitivity analysis of a 180-s CRDS flush followed by a 60-s average

High-resolution gas concentration mapping workflow

21 August 2026

Abstract

This reproducible sensitivity analysis reprocessed all four measurement campaigns using a 180-s CRDS line-flushing period followed by a 60-s concentration average. The average was assigned to the fourth minute of each nominal 240-s valve dwell and was harmonised with the existing FTIR observations. Means, medians, standard deviations, coefficients of variation (CV), block-wise leave-one-out relative errors (LOO RE), and spatial-temporal mixed models were re-calculated for CO₂, CH₄, NH₃, and three dimensionless gas ratios. Temporal two-hour blocks explained most of the modelled variation in CO₂ and CH₄. Spatial variation was more prominent for NH₃, particularly in Campaigns 2 and 3. Changing the CRDS averaging window altered campaign-level gas means by analyser-dependent amounts, with the largest median change observed for NH₃ in Campaign 1 and CH₄ in Campaign 2. The results are a processing sensitivity assessment and do not replace the primary manuscript analysis without a documented methodological decision.

1 Purpose and scope

The analysis tests the consequences of allowing three minutes for sample transport and line equilibration before retaining the final minute of every nominal four-minute CRDS valve position. It covers Campaigns 1–4. It also provides one harmonised data frame containing FTIR and CRDS observations with the established campaign-specific sampling-point mapping. Shannon entropy was deliberately excluded.

2 Data preparation

Per-second CRDS files were read from the recovered Picarro archives. Cyclic experimental periods were identified from the expected MPVPosition sequences. Within each accepted valve dwell, records from 0 to less than 180 s were discarded and records from 180 to less than 240 s were averaged. A minimum of 30 retained raw records and an observed dwell of at least 230 s were required. The timestamp was assigned to the whole minute containing the last retained record, representing the fourth minute of the dwell. Thus, the nominal interval remained 240 s.

FTIR observations were not re-averaged. Their native selector-level timestamps and Messstelle values were retained and sampling points were mapped using the same campaign metadata as the CRDS data. Campaign 1 FTIR values were harmonised to the CRDS scale by dividing CO₂ and CH₄ by 1.06 and NH₃ by 1.09. CRDS concentrations and H₂O were not corrected. The combined output preserves raw, corrected, and analysis-scale columns.

Only missing, zero, and negative response values were excluded response-wise from statistics. No IQR filter or other statistical outlier removal was applied. The background point `s` and the ring-line level `ring_in` remain in the combined data but were excluded from internal spatial comparisons.

Table 1: Prepared analyser observations. The number of sampling points includes retained background or ring-line levels, where applicable. The FTIR records have no CRDS raw-record count.

Campaign	Analyser	Observations	Levels	Median raw records
1	FTIR1	39,861	10	–
1	FTIR2	40,020	10	–
1	CRDS8	21,219	16	69
1	CRDS9	30,155	16	79
2	CRDS8	16,389	16	70
2	CRDS9	16,428	16	79
2	FTIR2	83	3	–
3	CRDS8	28,903	16	61
3	CRDS9	29,580	18	79
4	CRDS8	57,871	9	71

3 Statistical analysis

The arithmetic mean, median, standard deviation (SD), median absolute deviation, and CV were calculated for each response. The CV was defined as

$$\text{CV} = 100 \frac{\text{SD}(Y)}{\bar{Y}}.$$

The ratios NH_3/CO_2 , CH_4/CO_2 , and NH_3/CH_4 were calculated as dimensionless concentration ratios and were not multiplied by 100.

Sequential observations were assigned to clock-aligned two-hour blocks. A sampling-point median was calculated in each block. For sampling point i , the leave-one-out reference was the median of all other internal sampling points available in the same campaign, response, and block. The signed relative error was

$$\text{LOO RE}_{i,t} = 100 \frac{Y_{i,t} - \tilde{Y}_{-i,t}}{\tilde{Y}_{-i,t}}.$$

Heatmaps report the median absolute LOO RE over blocks.

Spatial and temporal contributions were evaluated with crossed random-intercept models fitted separately by campaign and response:

$$\log(Y_{i,t}) = \beta_0 + b_i + c_t + \varepsilon_{i,t},$$

where b_i is the sampling-point effect and c_t is the two-hour-block effect. Variance components are presented as percentages of total model variance. Vertical-group relevance was assessed for Campaigns 1–3 by likelihood-ratio comparison of models with and without vertical group as a fixed effect while retaining random intercepts for horizontal position and two-hour block.

4 Results

4.1 Network-level concentrations

Table 2: Network-level descriptive statistics after the 180-s flush and 60-s average. Concentrations are in ppm. CV is in percent.

Campaign	Gas	n	Mean	Median	SD	CV
1	CO ₂	127,267	622.77	612.26	106.94	17.17
1	CH ₄	127,270	18.76	18.02	8.10	43.19
1	NH ₃	127,189	2.09	2.01	0.77	36.85
2	CO ₂	32,873	636.80	628.11	118.27	18.57
2	CH ₄	32,873	18.37	17.47	9.67	52.66
2	NH ₃	32,873	0.86	0.79	0.39	45.75
3	CO ₂	55,187	559.36	535.69	99.98	17.87
3	CH ₄	55,187	12.11	10.37	6.95	57.36
3	NH ₃	55,187	0.70	0.61	0.39	56.35
4	CO ₂	45,010	565.68	547.14	84.48	14.93
4	CH ₄	45,010	12.37	11.04	6.43	51.97
4	NH ₃	45,010	0.60	0.55	0.30	49.89

The separation between means and medians was small for CO₂ but was larger for CH₄ and NH₃, consistent with right-skewed concentration distributions. CO₂ had the smallest network CV in every campaign. CH₄ and NH₃ were substantially more variable.

4.2 Effect of the changed CRDS dwell allocation

Table 3: Median analyser-level change in the campaign mean when the former 60-s flush plus 180-s average was replaced by a 180-s flush plus 60-s average. Values are percentages. Ranges show the minimum and maximum analyser-specific changes.

Campaign	Gas	Median change	Analyser range
1	CO ₂	0.95	−0.04 to 1.95
1	CH ₄	0.69	−0.13 to 1.50
1	NH ₃	5.33	−0.28 to 10.94
2	CO ₂	2.70	0.21 to 5.19
2	CH ₄	8.00	0.58 to 15.42
2	NH ₃	−1.72	−5.76 to 2.31
3	CO ₂	−1.52	−1.73 to −1.32
3	CH ₄	−5.34	−6.29 to −4.39
3	NH ₃	−4.34	−4.40 to −4.27
4	CO ₂	1.11	1.11 to 1.11
4	CH ₄	3.39	3.39 to 3.39
4	NH ₃	−0.37	−0.37 to −0.37

The direction and magnitude of the sensitivity were gas-, analyser-, and campaign-dependent. The comparison is not a strictly paired record-level experiment because recovered-file coverage and accepted-cycle coverage can differ between the old and new preparation workflows. It should therefore be interpreted as a pipeline-level sensitivity analysis.

4.3 Spatial and temporal variance

Table 4: Variance partition from crossed mixed models fitted to log-transformed two-hour sampling-point medians. Values are percentages of total model variance.

Campaign	Gas	Spatial SP	Temporal block	Residual
1	CO ₂	4.09	70.47	25.44
1	CH ₄	4.95	67.25	27.80
1	NH ₃	18.42	49.18	32.40
2	CO ₂	9.37	70.81	19.82
2	CH ₄	12.15	65.29	22.56
2	NH ₃	46.39	40.25	13.37
3	CO ₂	13.26	59.24	27.50
3	CH ₄	27.29	46.71	26.00
3	NH ₃	41.69	37.46	20.85
4	CO ₂	20.40	59.25	20.35
4	CH ₄	27.23	48.20	24.56
4	NH ₃	28.32	60.18	11.50

Temporal blocks accounted for 59.2–70.8% of CO₂ variance and 46.7–67.2% of CH₄ variance. NH₃ showed a stronger spatial contribution, reaching 46.4% in Campaign 2 and 41.7% in Campaign 3. All six vertical-group likelihood-ratio tests in each of Campaigns 1–3 were significant ($p < 10^{-8}$). These tests establish statistical vertical structure, but their large sample sizes mean that effect magnitude must be interpreted from the descriptive plots and variance components rather than from p -values alone.

5 Figures

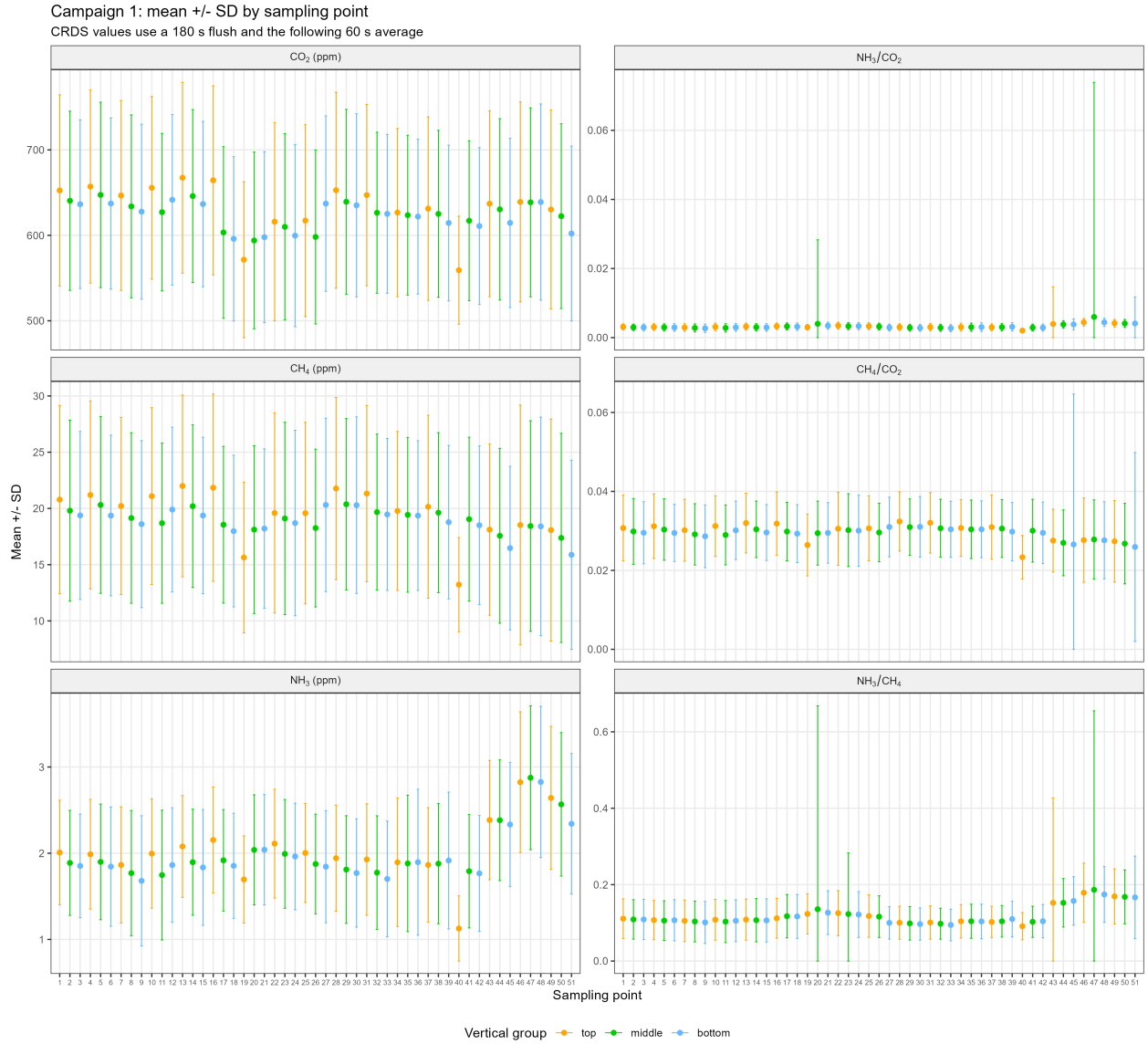


Figure 1: Campaign 1 mean $\pm$ SD for each gas and dimensionless ratio at each internal sampling point. Colours identify the top, middle, and bottom vertical groups.

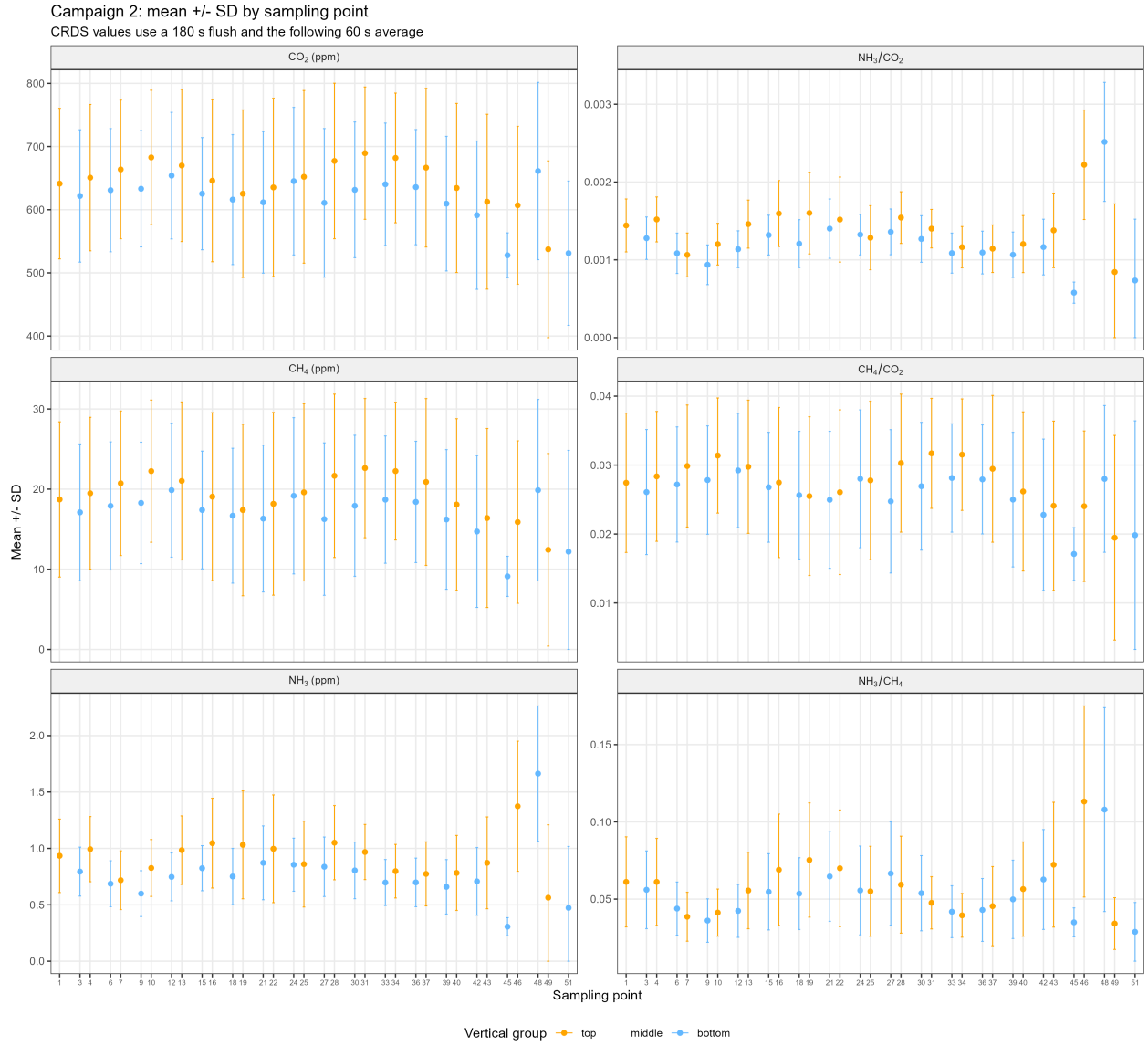


Figure 2: Campaign 2 mean $\pm$ SD by internal sampling point. The middle vertical group was not part of the CRDS configuration.

Campaign 3: mean $\pm$ SD by sampling point
 CRDS values use a 180 s flush and the following 60 s average

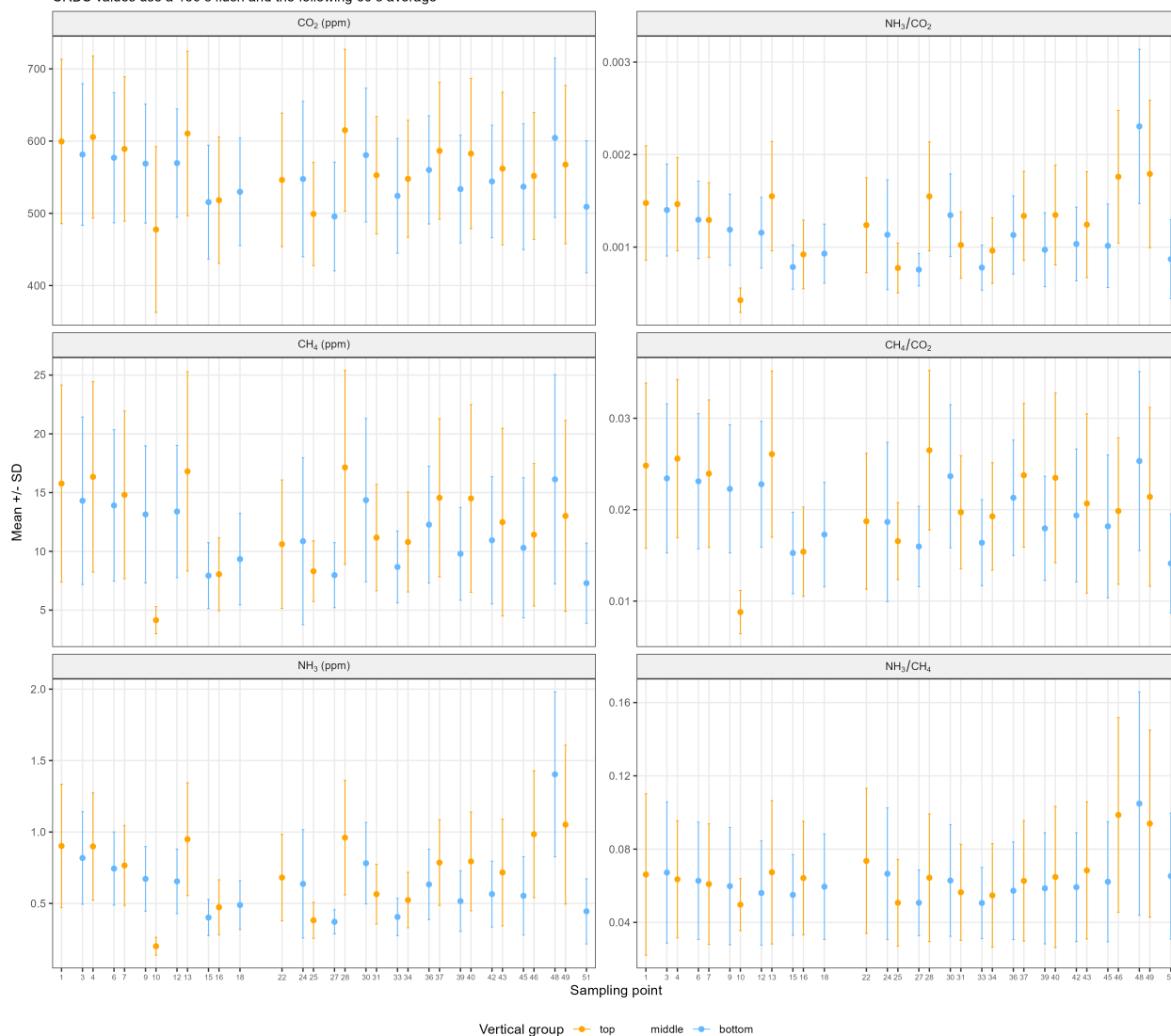


Figure 3: Campaign 3 mean $\pm$ SD by internal sampling point.

Campaign 4: mean $\pm$ SD by sampling point
 CRDS values use a 180 s flush and the following 60 s average

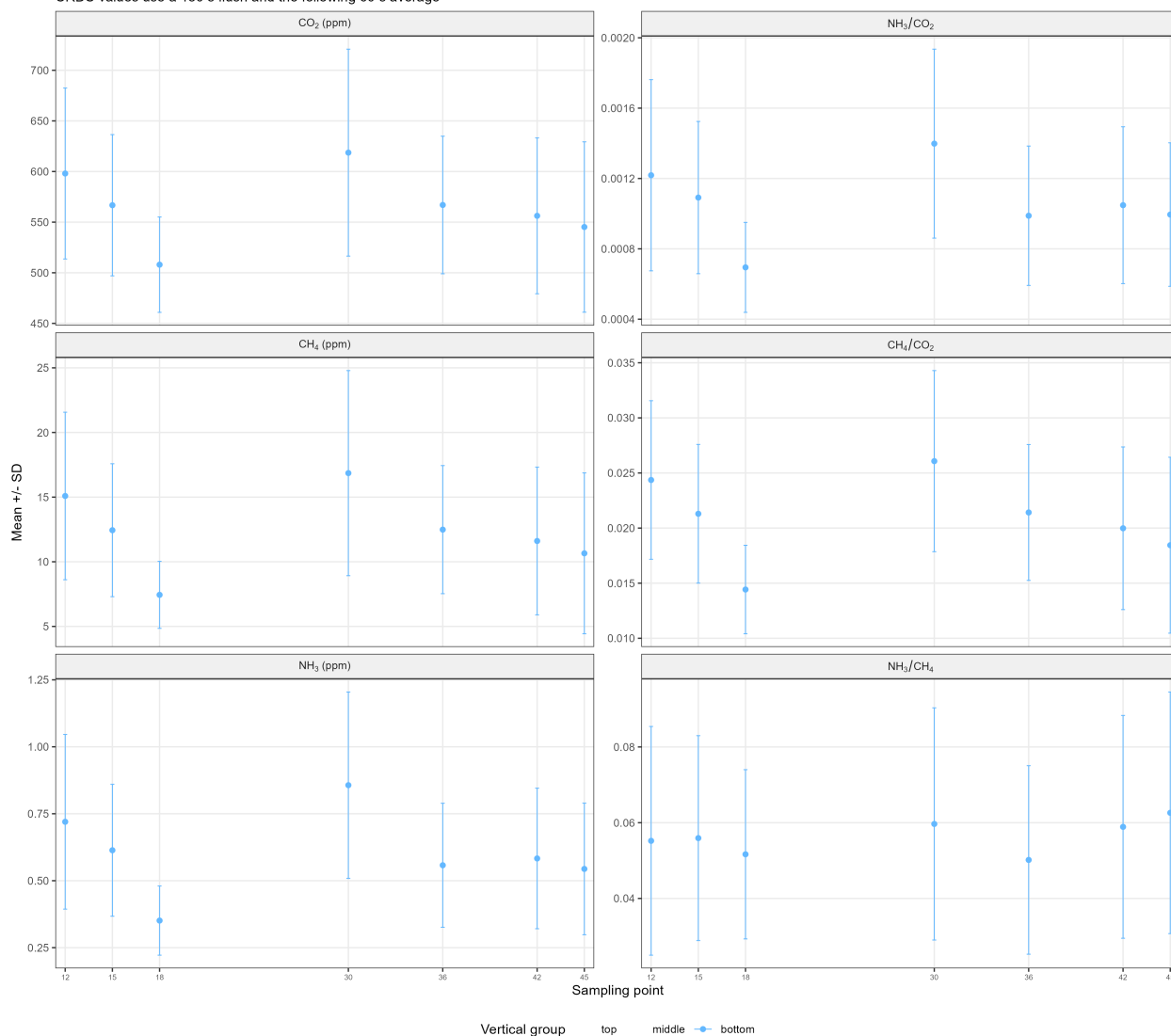


Figure 4: Campaign 4 mean $\pm$ SD by internal sampling point.

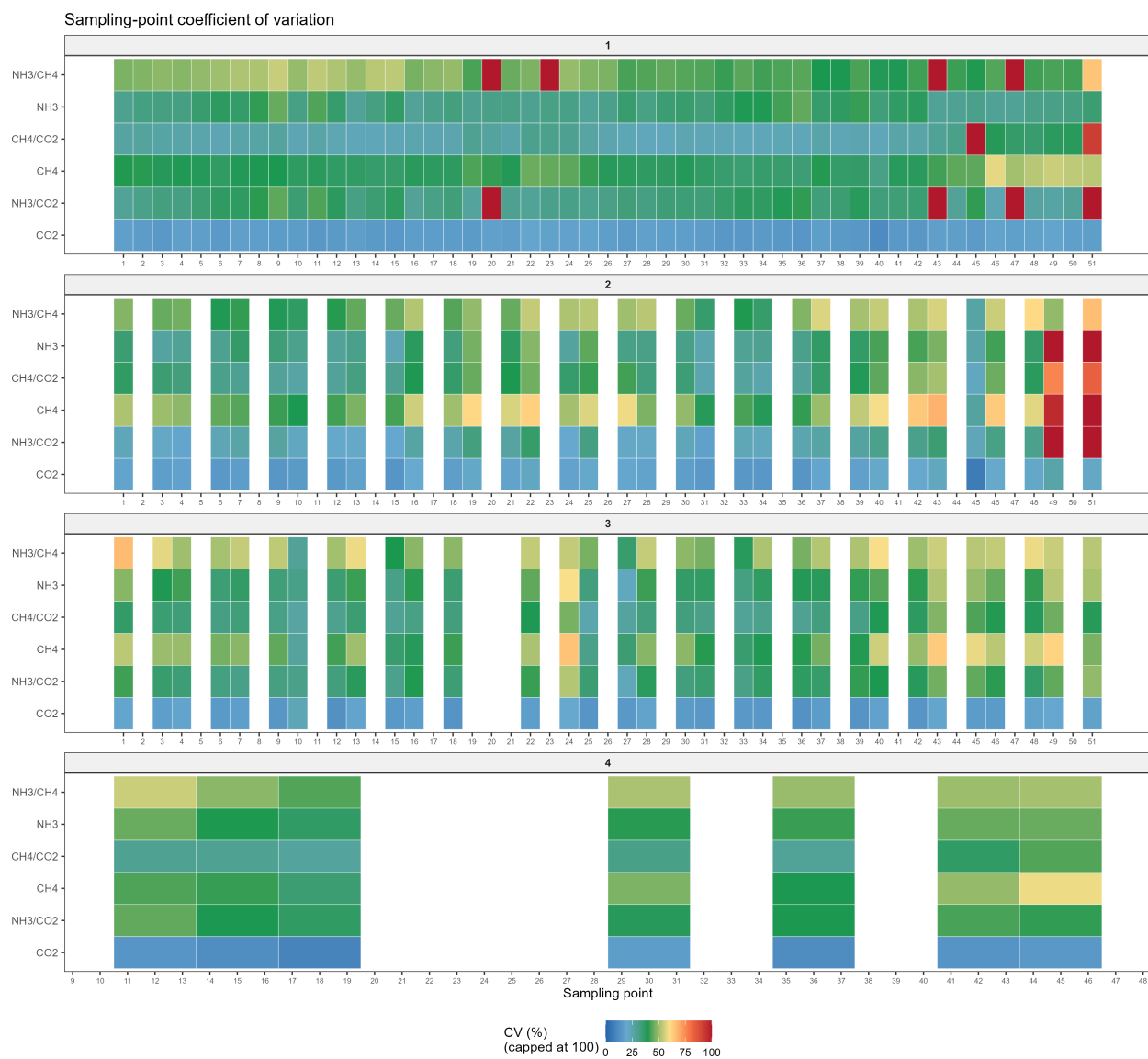


Figure 5: Sampling-point CV for all gases and ratios. Colour values are capped at 100% for display only. Uncapped values are retained in the CSV output. Blank cells indicate unmeasured sampling points.

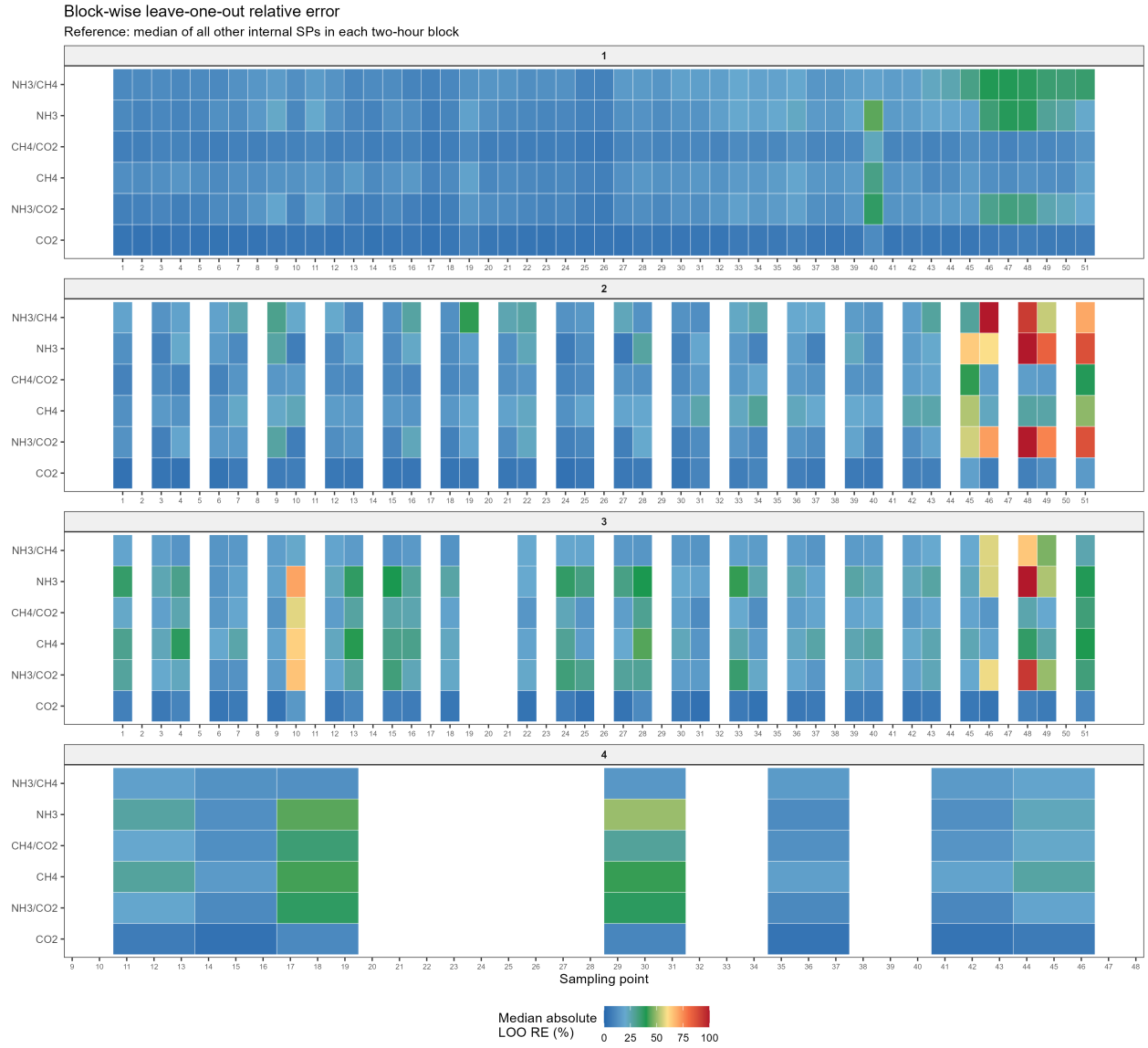


Figure 6: Median absolute leave-one-out relative error. Each sampling-point median was compared with the median of all other internal points available in the same two-hour block. Values are capped at 100% for display only.

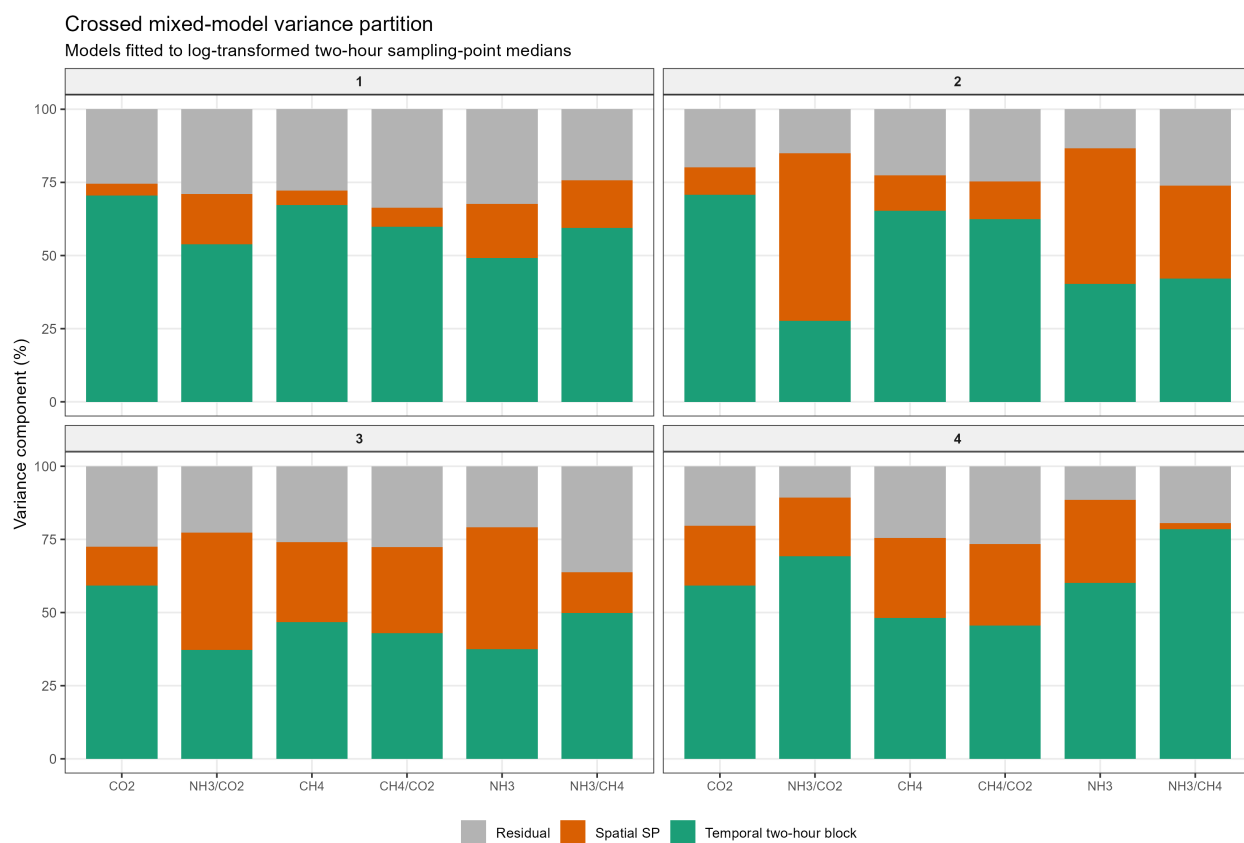


Figure 7: Spatial, temporal, and residual variance percentages from the crossed random-intercept models.

6 Interpretation and limitations

The 180-s flush plus 60-s averaging rule materially changes some campaign-level estimates, particularly CH₄ in Campaign 2 and NH₃ in Campaign 1. A final decision should consider physical residence time, analyser response, line length, and stability during the retained minute. The shortened average also increases sensitivity to short-lived peaks.

All positive concentrations were retained as requested. Consequently, very small positive denominator concentrations can produce extreme gas ratios and large ratio SD or CV values. These records are visible in the outputs rather than being hidden by a post-processing threshold. Before using ratios for inference, their corresponding numerator and denominator time series should be reviewed and a physically justified quality rule should be specified prospectively.

Campaign 3 positions that changed function over time were mapped according to the documented switch date. Campaign 4 contains seven internal points plus non-internal line levels. Campaign 2 has sparse FTIR2 coverage. These differences constrain direct comparisons of network density between campaigns. The variance partitions describe the observed data hierarchy. They do not identify causal mechanisms or adjust for weather, management, or animal activity.

7 Reproducibility

The raw-data preparation is implemented in `scripts/04_prepare_crds_180s_flush_60s_average_all_campaigns.R`. Statistical tables and figures are produced by `scripts/05_crds_180s_flush_60s_statistics_plots.R`. Prepared measurement CSV files are intentionally excluded from Git. The R scripts, this report source, the compiled PDF, and the plotted PNG files are version controlled.